

20CS7T02 – Cloud Computing

UNIT 1: Cloud Fundamentals

Introduction

Cloud computing provides on-demand access to computing resources over the internet. It eliminates the need for local infrastructure and enables scalable and flexible computing.

This unit introduces cloud concepts and characteristics.

Detailed Explanation

Characteristics of Cloud Computing

Key features include scalability, elasticity, and pay-as-you-go pricing.

Users can access resources anytime and anywhere.

Benefits of Cloud Computing

Cloud computing reduces costs and improves efficiency.

Applications

Used in web hosting, data storage, and enterprise applications.

Summary

This unit introduces cloud fundamentals.

UNIT 2: Virtualization

Introduction

Virtualization is the process of creating virtual versions of physical resources.

Detailed Explanation

Types of Virtualization

Includes server, storage, and network virtualization.

Hypervisors

Manage virtual machines.

Applications

Used in cloud infrastructure.

Summary

This unit explains virtualization.

UNIT 3: Cloud Services Models

Introduction

Cloud services are categorized into different models.

Detailed Explanation

IaaS, PaaS, SaaS

Each model provides different levels of control.

Applications

Used in cloud services.

Summary

This unit explains service models.

UNIT 4: Cloud Deployment Models

Introduction

Cloud deployment models define how cloud services are deployed.

Detailed Explanation

Types

Public, private, hybrid clouds.

Applications

Used in organizations.

Summary

This unit explains deployment models.

UNIT 5: Cloud Security

Introduction

Cloud security ensures data protection.

Detailed Explanation

Threats and Solutions

Includes encryption and access control.

Applications

Used in secure cloud systems.

Summary

This unit explains cloud security.